

**The Big Idea: Law of Conservation of Energy** S8P2

Energy cannot be created or destroyed — it can only be **transformed** from one form to another or **transferred** from one object to another. The total energy in a closed system always stays the same!

$$\text{Total Energy IN} = \text{Total Energy OUT} \quad \checkmark$$



Georgia connection: This is why a roller coaster at **Six Flags Over Georgia** doesn't need a motor after the first hill — the energy just keeps transforming between different forms as the car moves!

**Forms of Energy** S8P2c

Every form of energy falls into two main categories: **POTENTIAL** (stored) or **KINETIC** (motion). Here are the most common forms you'll see on the Milestones:

**Mechanical**

Motion of objects (a rolling car, spinning wheel)

**Thermal (Heat)**

Energy of moving particles (hot pan, fire)

**Light (Radiant)**

Electromagnetic energy (sun, lamp, screen)

**Electrical**

Flow of electrons (outlets, batteries, lightning)

**Sound**

Vibrations through a medium (speakers, voices)

**Chemical**

Stored in bonds between atoms (food, fuel, batteries)

**Nuclear**

Stored in atomic nuclei (nuclear plants, the Sun)

**Gravitational**

PE due to height above the ground (a book on a shelf)

**Kinetic & Potential Energy Relationships** S8P2a**KINETIC ENERGY (KE)**

Energy of motion. Depends on two things:

- **Mass** — heavier = more KE
- **Speed** — faster = more KE (even bigger effect!)

$$KE = \frac{1}{2} \times m \times v^2$$

Example: A fast-moving bowling ball has more KE than a slow-moving marble.

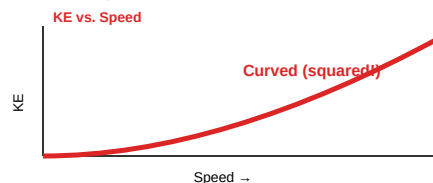
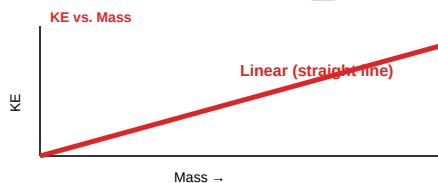
**POTENTIAL ENERGY (PE)**

Stored energy due to position. Depends on:

- **Mass** — heavier = more PE
- **Height** — higher = more PE

$$PE = m \times g \times h$$

Example: A book on a high shelf has more PE than the same book on the floor.

**How KE changes with Mass & Speed**

Key insight: Doubling the **mass** → doubles the KE. But doubling the **speed** → QUADRUPLES the KE (because speed is squared)!

**KE ↔ PE Transformations** S8P2b

On a roller coaster, pendulum, or rubber band, energy keeps swapping between KE and PE. The **total** stays the same (Conservation of Energy)!



At the TOP of the hill: All PE, no KE (car is still)

Going DOWN: PE → KE (car speeds up!)

At the BOTTOM: All KE, no PE

Going UP the next hill: KE → PE

Other examples: Pendulum (clock), bouncing ball, stretched rubber band, bow & arrow

**ACTIVITY 1: KE & PE Practice**

1. Which has more KE: a 2 kg ball moving at 3 m/s OR a 2 kg ball moving at 6 m/s? _____ Why? _____
2. Stone Mountain is about 1,686 feet tall. Which has more PE: a person standing at the base or a person at the top? _____
3. A pendulum swings back and forth. At the highest point of the swing, the pendulum has the MOST _____ energy and the LEAST _____ energy.
4. A rubber band is stretched and held still. What kind of energy does it have? _____

⚡ Flip over for energy transformations in systems + heat transfer →

Energy Transformations in Systems S8P2c

Any time energy changes form, it's a **transformation**. When you make a claim about what kind of transformation is happening, back it up with **evidence** — what you observe!

Everyday Example	Transformation	Evidence
Lighting a match	Chemical → Light + Heat (Thermal)	Flame glows; you feel warmth
Turning on a lamp	Electrical → Light + Heat	Bulb glows; bulb gets warm
Eating breakfast & running	Chemical → Mechanical + Thermal	You move, you sweat
Playing a radio	Electrical → Sound	You hear music
Solar panel charging a phone	Light → Electrical → Chemical (in battery)	Phone battery fills up
Hydroelectric dam (water to power)	PE (water up high) → KE (falling water) → Mechanical (turbine) → Electrical	Lights turn on in homes

GEORGIA CONNECTION President FDR visited **Warm Springs, GA** for his polio treatment. Georgia's **Rural Electrification Administration (New Deal)** brought electricity to GA farms in the 1930s — hydroelectric dams on GA rivers transformed **PE → KE → Mechanical → Electrical** energy so rural families could turn on lights for the first time! (SS8H8e)

Heat Transfer & Molecular Motion S8P2d

Heat moves from **HOT** to **COLD**. When particles gain thermal energy, they move **faster and farther apart**. When they lose it, they move slower. Heat transfers in three ways:

CONDUCTION

Through direct contact. Atoms/molecules collide and transfer energy.

Best in: solids (especially metals)

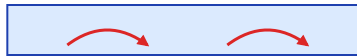


Example: Spoon gets hot in soup; iron cools by contact with ice

CONVECTION

Through currents in fluids (liquids & gases). Hot fluid rises; cold sinks.

Best in: liquids and gases



Example: Boiling water; warm air rising in a room; ocean currents

RADIATION

Through electromagnetic waves. Needs **NO** matter — works in space!

Best in: empty space or through air



Example: Sun heats Earth; fire warms you from across the room; microwave oven

GEORGIA CONNECTION On a hot summer day in Atlanta, you experience ALL THREE: **Radiation** from the sun heats the I-75 asphalt → **Conduction** sends heat up through your shoe soles → **Convection** rises in the air above the road, creating those shimmering "heat waves" you see over highways. (SS8E1a — Interstate Highway System)

Bonus: Energy Transformations in Georgia History SS connection

Bell Bomber Plant (WWII) (SS8H9b)

Built B-29 bombers in Marietta. The engines transformed:

Chemical (fuel) → Thermal → Mechanical → KE

Atlanta Railroads (SS8H4c)

Steam locomotives transformed:

Chemical (coal) → Thermal (steam) → Mechanical (wheels) → KE

ACTIVITY 2: Identify the Transformation

Name the main energy transformation for each:

1. A firefly glowing → _____ → _____
2. A toaster heating bread → _____ → _____
3. A hydroelectric dam → _____ → _____
4. A campfire → _____ → _____ + _____
5. A ceiling fan → _____ → _____

ACTIVITY 3: Heat Transfer — Pick One

Label each as **CD** (conduction), **CV** (convection), or **R** (radiation):

1. _____ Warming your hands over a campfire
2. _____ A metal spoon in a bowl of hot soup
3. _____ Hot air rising from a heater
4. _____ Sun heating your face at Tybee Beach
5. _____ A pot of water boiling on the stove
6. _____ Ice cube melting in your hand

I Can... (Self-Check for Milestones)

- | | |
|---|--|
| <ul style="list-style-type: none"> ☐ State the Law of Conservation of Energy ☐ Identify 8 forms of energy (mechanical, thermal, light, electrical, sound, chemical, nuclear, gravitational) ☐ Explain how KE depends on mass and speed ☐ Explain how PE depends on mass and height ☐ Describe KE/PE transformations (roller coaster, pendulum, rubber band) | <ul style="list-style-type: none"> ☐ Claim & support a transformation (e.g., match = chemical → light + heat) ☐ Define conduction, convection, and radiation ☐ Match each heat transfer type to a real example ☐ Explain how heat transfer relates to molecular motion ☐ Apply energy ideas to Georgia examples (Warm Springs, Bell Bomber, I-75, railroads) |
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